Password Strength Analyzer

IBM SkillsBuild Cyber Security BootCamp 2026 | BharatCares

Project Description

A Python tool that measures how strong a password is. It calculates the password's **entropy**, checks it against a list of **common passwords**, detects weak patterns (sequences like abc/123 and repeated characters), and returns a **score out of 100**, a **rating** (Very Weak to Very Strong) and **suggestions** to improve it. It also includes a **secure password generator** and a bar-chart comparison of multiple passwords.

Why It Matters

Weak passwords are one of the most common causes of account breaches. Attackers use dictionary and brute-force attacks, so a password's length, variety and unpredictability directly affect how long it would take to crack.

Technologies Used

- Python 3
- Jupyter Notebook / Google Colab
- pandas (results table)
- `matplotlib` (visualization)
- Standard library: `math`, `re`, `string`, `secrets`, `getpass`

Dataset

No external dataset is required. The project uses a small built-in list of common passwords inside the notebook. For a larger wordlist (optional upgrade), the public RockYou list can be used: https://github.com/brannondorsey/naive-hashcat/releases/download/data/rockyou.txt

Project Files

File	Description
`Dipanksha_Elinje_PasswordStrengthAnalyzer.ipynb`	Complete project code
`requirements.txt`	Python dependencies
`Dipanksha_Elinje_ProjectReport.docx`	Full project report
`README.md`	This file

Setup and Run

Option 1: Google Colab (easiest) 1. Go to https://colab.research.google.com and choose *File* > *Upload notebook*. 2. Upload `Dipanksha_Elinje_PasswordStrengthAnalyzer.ipynb`. 3. Click *Runtime* > *Run all*.

Option 2: Local machine

```
pip install -r requirements.txt
jupyter notebook Dipanksha_Elinje_PasswordStrengthAnalyzer.ipynb
```

Then run all cells.

How It Works

1. **Entropy** = length x log2(character pool size)
2. **Score** = entropy-based points + length bonus + character-variety bonus - penalties
3. **Penalties** for common passwords, sequences and repeated characters
4. **Rating**: <30 Very Weak, <50 Weak, <70 Moderate, <85 Strong, 85+ Very Strong

Sample Output

Password	Score	Rating
`123456`	0	Very Weak
`hello2026`	56	Moderate
`Hello@2026`	80	Strong
`T!g3r-Blue-Moon-88`	100	Very Strong

Key Information

- Passwords entered are never stored or sent anywhere.
- The generator uses Python's `secrets` module (cryptographically secure).

Future Improvements

- Check passwords against the Have I Been Pwned breach database
- Streamlit web interface
- ML model trained on a leaked-password dataset

Author

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